1-Deck Casino Card to BIP-39 Seed Record

Generator version 2.12, derivation id "1deck-bip39-v2".
Generated offline from physical card draws. No random number generator was used at any point: every entropy bit below comes from cards that were shuffled, drawn and typed in by hand.

THIS DOCUMENT IS YOUR SEED

Anyone holding this file can spend your funds. The card draws alone reproduce the seed, so redacting the word list does not make it safe. Print it and destroy the digital copy, or store it exactly as you would store the seed phrase itself. Never place it on an internet-connected or cloud-synced machine.

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1. CONFIGURATION

Seed length	`12 words (128-bit entropy)`
Protocol	`Reshuffle the full deck between every draw`
Entropy setting	`Safety margin`
Physical draws	`8 draws of 4 cards = 32 entries`
Deck	`one standard 52-card deck, reshuffled between draws`

2. ENTROPY ACCOUNTING

Card entropy available	`181.05 bits`
Bits extracted	`128 bits`
Head-room	`53.05 bits`
Extraction efficiency	`70.7 %`

Head-room is the surplus that lets SHA-256 act as a randomness extractor: it is what absorbs an imperfect shuffle instead of passing the bias through to the seed.

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3. PHYSICAL DRAWS, IN THE ORDER THE CARDS CAME OFF THE DECK

Ranks A 2 3 4 5 6 7 8 9 T J Q K   Suits C=Clubs D=Diamonds H=Hearts S=Spades. Order matters: the same four cards in a different order give a different seed.

```
Draw  1 :  TD  KD  8H  5C    (10 of Diamonds, King of Diamonds, 8 of Hearts, 5 of Clubs)
Draw  2 :  TS  2D  8C  JH    (10 of Spades, 2 of Diamonds, 8 of Clubs, Jack of Hearts)
Draw  3 :  8C  9C  KH  TC    (8 of Clubs, 9 of Clubs, King of Hearts, 10 of Clubs)
Draw  4 :  KD  7H  3S  QH    (King of Diamonds, 7 of Hearts, 3 of Spades, Queen of Hearts)
Draw  5 :  8H  7H  AD  JH    (8 of Hearts, 7 of Hearts, Ace of Diamonds, Jack of Hearts)
Draw  6 :  7S  KS  5S  6C    (7 of Spades, King of Spades, 5 of Spades, 6 of Clubs)
Draw  7 :  5D  QH  7S  3C    (5 of Diamonds, Queen of Hearts, 7 of Spades, 3 of Clubs)
Draw  8 :  AC  JS  QS  3H    (Ace of Clubs, Jack of Spades, Queen of Spades, 3 of Hearts)
```

SHUFFLE CHECK

No sign of under-shuffling: 4 adjacent pairs, against an expected 1.7.
Counted: the 48 pairs of cards dealt within 3 positions of each other that share a suit and sit one rank apart. A sealed deck arrives with each suit in rank order, so surviving pairs are the fingerprint of a deck that was not riffled enough. Expected 1.74 for a well-shuffled deck; warning above 5, strong alarm above 7.
Pairs found: 8C and 9C (draw 3); 9C and TC (draw 3); 8H and 7H (draw 5); JS and QS (draw 8)
This is a blunder detector, not a certificate: it catches one or two riffles reliably, is blind to four or more, and assumes the deck started in factory order. A deck already in some other known order - a cancelled casino deck resealed in a box, for instance - passes this check regardless.

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4. DERIVATION - REPRODUCIBLE WITH ANY SHA-256 TOOL

Step 1. Every card, in order, is written into one domain-separated ASCII string:

```
1deck-bip39-v2|12|TD KD 8H 5C TS 2D 8C JH 8C 9C KH TC KD 7H 3S QH 8H 7H AD JH 7S KS 5S 6C 5D QH 7S
3C AC JS QS 3H
```

Step 2. That string is hashed. Reproduce it on any command line with the following, which is ONE line - if it wraps below, join it back together before running it:

```
printf '%s' '1deck-bip39-v2|12|TD KD 8H 5C TS 2D 8C JH 8C 9C KH TC KD 7H 3S QH 8H 7H AD JH 7S KS 5S
6C 5D QH 7S 3C AC JS QS 3H' | sha256sum
```

SHA-256 of preimage

`56cd7feb1647fd133477bc54589f7c2d5eefb33dec0fac06f085518495a628a2`

Step 3. The leading 128 bits of that digest are the entropy:

```
56cd7feb1647fd133477bc54589f7c2d
```

Step 4. The BIP-39 checksum is computed from that entropy - it is not drawn and not chosen. SHA-256(entropy) = 8d228cb19edc18fe9b18551bafbe5e0d32d45f211d3a82f0c5416b83889769ba; its leading 4 bits (1000) are appended, and the resulting 132 bits are split into 12 groups of 11, each indexing the official BIP-39 English word list.

Seed Phrase

SECRET - ANYONE WHO READS THIS CAN SPEND YOUR FUNDS

```
1. fine           5. lemon          9. february
2. hip            6. maze          10. shaft
3. width          7. spike         11. tenant
4. clutch         8. wasp          12. force
```

FULL SEED PHRASE

`fine hip width clutch lemon maze spike wasp february shaft tenant force`

VERIFICATION

BIP-39 checksum `VALID - re-checked against the BIP-39 rules`

First receive addresses `native SegWit, BIP-84`

```
m/84'/0'/0'/0/0   bc1qv96hhcapy3x55jk8w59nsxeut5vytnuga8fn6l
m/84'/0'/0'/0/1   bc1qp3phvf9aagm6t74geaspmcpy3vac9h0mt827my
```

Master fingerprint `B8428EFE   (no passphrase)`

Restore these words into your wallet or signing device and compare the fingerprint it shows. If the eight characters match, every word arrived intact. A BIP-39 passphrase changes this value, and is deliberately not recorded here.

Draw fingerprint

`9deae0f02fc04eb9fb5f5ea3b4597ce0b6ca61e2883b945772ffe79c044de21f`

To confirm this record is intact, re-enter the draws from section 3 into the generator and check that you get the same seed phrase. The fingerprint above is SHA-256 over the string "fingerprint|" followed by the card codes, and is what the tool's verify-by-re-entry mode compares.

No date is recorded in this file. The generator has no access to the system clock by design, so that nothing time-derived can enter the derivation.

RECOVERY WITHOUT THIS GENERATOR

The entropy in section 4 is the universal interchange format: any BIP-39 tool can turn it back into these exact words.

```
1.  Entropy:  56cd7feb1647fd133477bc54589f7c2d
     or recompute it from the draws in section 3, as shown there.
2.  Open a BIP-39 tool. iancoleman.io/bip39 is the most widely mirrored;
     use its offline standalone copy, on a disconnected machine.
3.  Set  Entropy Type = Hex.  Set it explicitly - auto-detect can read
     an all-digit entropy as base 10 and silently give the wrong seed.
4.  Set  Mnemonic Length = "Use Raw Entropy".  NOT "12 Words" - that
     hashes your entropy a second time and yields a different seed.
5.  Paste the entropy. The phrase appears immediately.
6.  Verify: BIP84 tab, first address = bc1qv96hhcapy3x55jk8w59nsxeut5vytnuga8fn6l
```

Do not paste the CARDS into another tool and expect these words: other tools encode cards differently, so you would get a valid but different seed. The entropy hex is what they share.

STORAGE

Store on paper or metal, offline, where only you can reach it. Test recovery with a small amount before committing real funds. A BIP-39 passphrase is a second factor and is not recorded here.